Tristan Bullion-Gauthier

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Education

2024- **Mathematics PhD**, *Institut Camille Jordan, Université Claude Bernard Lyon 1*, Lyon, France
Under the supervision of Óscar Domínguez and Petru Mironescu.

2023-2024 **M2 in Advanced Mathematics, PDE and applications**, *École Normale Supérieure de Lyon / Université Claude Bernard Lyon 1*, Lyon, France.
PDE and applications
Research internship on *Interpolation theory and function spaces*, under the supervision of Petru Mironescu.

2022-2023 **M2 Mathématiques Générales**, *Université Claude Bernard Lyon 1*, Lyon, France.
Agrégation, rank 60.

2021-2022 **M1 Mathématiques Générales**, *Université Claude Bernard Lyon 1*, Lyon, France.

2020-2021 **L3 Mathématiques Générales**, *Université Claude Bernard Lyon 1*, Lyon, France.

2018-2020 **CPGE MPSI / MP***, *Lycée du Parc*, Lyon, France.

Publications

1. **Higher-order affine Sobolev inequalities**, *published in the Journal of Functional Analysis*, 2026
2. **Existence of extremal functions in higher-order affine Sobolev inequalities**, *preprint*, 2026
3. L^2 **regularity of minimizers of anistropic interaction functionals**, *with Xiao Kai*, preprint, 2026

Teaching

2026-2027 **Exercise sessions Techniques mathématiques de base (21 hours)**

2025-2026 **Exercise sessions Analysis 2 (36 hours)**, *Université Claude Bernard Lyon 1.*

2025-2026 **Exercise sessions Mathematics 3 (36 hours)**, *Université Claude Bernard Lyon 1.*

2024-2025 **Exercise sessions Analysis 2 (36 hours)**, *Université Claude Bernard Lyon 1.*

2024-2025 **Oral interrogations, first year**, *Université Claude Bernard Lyon 1.*

Talks

Functional inequalities seminar, *Institut Camille Jordan*, Lyon.

Phd students seminar, *Institut Camille Jordan*, Lyon.

BARcamp, Initiation à la recherche, *BU Sciences Lyon 1*
Outreach presentation.

InfoMaths Doctoral school day, *Université Claude Bernard Lyon 1*, Lyon.

Other activities

2025 - 2026 Member of the council of the Lyon 1 mathematics department